

Term 3 — Lesson 1 Practice Activity 2 (C1–C3) with Solutions

Contents

- 1 Introduction
- 2 Math-support program and average algebra score (lower-tailed direction)
- 3 Delivery app on-time rate after route training (higher-tailed direction)
- 4 Clinic waiting time consistency check (two-tailed direction)
- 5 Fuel efficiency review under different policy strictness (lower tail, changing α)
- 6 Scholarship completion-rate target (higher tail, changing α)
- 7 Factory packaging-weight stability check (two tail, changing α)
- 8 App response time: lower vs different on the same data
- 9 Daily study-time claim: higher vs different on the same data

Evaluated criteria

- C1: Describe the concept of a statistical hypothesis in context.
- C2: Identify the null and alternative hypotheses using correct parameter notation.
- C3: Determine whether a test is one tailed or two tailed based on the claim.

1 Introduction

This activity is organized as a progression. In Block 1, you identify the correct test direction from context and complete the test decision. In Block 2, the hypotheses and tail direction remain fixed while the significance level changes, so you can see how conclusions can change when standards of evidence change. In Block 3, you compare one-sided and two-sided alternatives on the same evidence.

Use statistically precise language in your conclusions: *reject H_0* or *fail to reject H_0* .

2 Math-support program and average algebra score (lower-tailed direction)

A school mathematics department introduced a new support schedule for algebra and wants to confirm that the change has not unintentionally lowered performance. The department uses seventy two points as its established benchmark for the usual quiz average in this course, so that value is the reference for the review. To evaluate the new schedule, staff collected results from thirty six students and found an average of sixty nine point five points. From past records they treat the typical spread in scores as about nine points, and they will judge the evidence at a five percent significance level.

Questions/Tasks.

(C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Mean setting. Let μ be the true mean algebra score; $\mu_0 = 72$, $n = 36$, $\bar{x} = 69.5$, $\sigma = 9$, $\alpha = 0.05$.

C2

$H_0 : \mu = 72$, $H_a : \mu < 72$.

C3

Left-tailed test because the claim is “reduced” (lower than benchmark).



Extra analysis

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{69.5 - 72}{9/6} = \frac{-2.5}{1.5} = -1.67.$$

For a left tail, $p \approx P(Z \leq -1.67) \approx 0.048$. Since $0.048 < 0.05$, reject H_0 .

Conclusion: At the 5% level, there is sufficient evidence that the true mean algebra score is below 72.

3 Delivery app on-time rate after route training (higher-tailed direction)

A local delivery company invested in new route training and management now wants evidence that service reliability is genuinely better than the long run standard they have reported for years. Their historical benchmark is an on

time rate of eighty percent, and the current claim is that performance has moved higher than that level. To test this, the operations team reviewed two hundred fifty recent deliveries and recorded that two hundred fourteen were on time. They will evaluate the result using a five percent significance level.

Questions/Tasks.

- (C1) Identify parameter type and symbols.
- (C2) Write the context-appropriate hypotheses.
- (C3) Classify the tail and justify it.
- (EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Proportion setting. Let p be true on-time proportion; $p_0 = 0.80$, $n = 250$, $x = 214$, $\hat{p} = 214/250 = 0.856$, $\alpha = 0.05$.

C2

$H_0 : p = 0.80$, $H_a : p > 0.80$.

C3

Right-tailed test because claim is “increased above 80%.”



Extra analysis

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.856 - 0.80}{\sqrt{0.8(0.2)/250}} = \frac{0.056}{0.0253} \approx 2.21.$$

Right-tail p-value: $p \approx P(Z \geq 2.21) \approx 0.014$. Since $0.014 < 0.05$, reject H_0 .

Conclusion: At the 5% level, evidence supports that the true on-time proportion is above 80%.

4 Clinic waiting time consistency check (two-tailed direction)

A school clinic recently changed staffing and administrators need to know whether patient flow has shifted away from the service level families are used to. The clinic benchmark is an average wait of eighteen minutes, and the concern is any meaningful change rather than only an increase or only a decrease. To check this, staff sampled forty nine visits and observed an average waiting time of eighteen point nine minutes. Historical monitoring indicates a typical variation of about four point two minutes, and the decision will be made at a five percent significance level.

Questions/Tasks.

- (C1) Identify parameter type and symbols.
- (C2) Write the hypothesis pair that fits the context.
- (C3) Classify the tail and justify it.
- (EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Mean setting. Let μ be true mean waiting time; $\mu_0 = 18$, $n = 49$, $\bar{x} = 18.9$, $\sigma = 4.2$, $\alpha = 0.05$.

C2

$$H_0 : \mu = 18, H_a : \mu \neq 18.$$

C3

Two-tailed test because “different” allows either direction.



Extra analysis

$$z = \frac{18.9 - 18}{4.2/\sqrt{49}} = \frac{0.9}{0.6} = 1.50.$$

Two-tailed p-value: $p \approx 2P(Z \geq 1.50) \approx 0.134$. Since $0.134 > 0.05$, fail to reject H_0 .

Conclusion: At the 5% level, evidence is insufficient to conclude the true mean waiting time is different from 18 minutes.

5 Fuel efficiency review under different policy strictness (lower tail, changing α)

A transport office introduced a new maintenance cycle and needs to verify that it has not reduced fleet fuel efficiency below the operating standard used for budgeting and planning. The office benchmark is fourteen point zero kilometers per liter for the average bus, so the investigation focuses on whether actual performance has dropped under that target. Inspectors sampled sixty four buses and found an average of thirteen point seven zero kilometers per liter. Long term records support using one point two kilometers per liter as the known standard deviation. For routine monitoring the office applies a ten percent significance level, while annual accountability reports use a five percent significance level.

Questions/Tasks.

- (C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Mean setting. μ : true mean fuel efficiency; $\mu_0 = 14.0$, $n = 64$, $\bar{x} = 13.70$, $\sigma = 1.2$.

C2

C2 (fixed across both levels). $H_0 : \mu = 14.0$, $H_a : \mu < 14.0$.

C3

C3 (fixed across both levels). Left-tailed test.



Extra analysis

$$z = \frac{13.70 - 14.0}{1.2/\sqrt{64}} = \frac{-0.30}{0.15} = -2.00, \quad p \approx P(Z \leq -2.00) = 0.0228.$$

- At $\alpha = 0.10$: $0.0228 < 0.10$ so reject H_0 .
- At $\alpha = 0.05$: $0.0228 < 0.05$ so reject H_0 .

To show sensitivity, compare with a stricter audit level $\alpha = 0.01$: $0.0228 > 0.01$, so fail to reject H_0 .

Conclusion: Hypotheses and tail type remain fixed; the decision changes only when the evidence standard becomes much stricter.

6 Scholarship completion-rate target (higher tail, changing α)

A scholarship foundation funded a mentoring program and now needs to show whether completion outcomes are truly better than the baseline used in its planning documents. The benchmark completion rate is sixty percent, and leadership is specifically looking for evidence of improvement above that level before expanding funding. The evaluation team reviewed two hundred students in the program and counted one hundred thirty one completions. Internal planning decisions use a ten percent significance level, while public reporting requires a five percent significance level.

Questions/Tasks.

(C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Proportion setting. p : true completion proportion; $p_0 = 0.60$, $n = 200$, $x = 131$, $\hat{p} = 0.655$.

C2

C2 (fixed). $H_0 : p = 0.60$, $H_a : p > 0.60$.

C3

C3 (fixed). Right-tailed test.



Extra analysis

$$z = \frac{0.655 - 0.60}{\sqrt{0.6(0.4)/200}} = \frac{0.055}{0.0346} \approx 1.59, \quad p \approx P(Z \geq 1.59) \approx 0.056.$$

- At $\alpha = 0.10$: $0.056 < 0.10$ so reject H_0 .
- At $\alpha = 0.05$: $0.056 > 0.05$ so fail to reject H_0 .

Conclusion: Same hypotheses and right-tail direction, but the final decision changes because α changes.

7 Factory packaging-weight stability check (two tail, changing α)

A factory quality team monitors package weight to ensure products stay aligned with the official specification used for customer consistency and regulatory compliance. The reference value is five hundred grams, and supervisors want to detect any shift away from that target in either direction. During inspection they sampled one hundred packages and obtained an average weight of five hundred one point two grams. Process history provides a known standard deviation of six grams. Routine internal checks use a ten percent significance level, and regulatory reporting uses a five percent significance level.

Questions/Tasks.

(C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

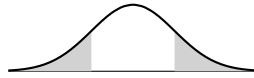
Mean setting. μ : true mean package weight; $\mu_0 = 500$, $n = 100$, $\bar{x} = 501.2$, $\sigma = 6$.

C2

C2 (fixed). $H_0 : \mu = 500$, $H_a : \mu \neq 500$.

C3

C3 (fixed). Two-tailed test.



Extra analysis

$$z = \frac{501.2 - 500}{6/\sqrt{100}} = \frac{1.2}{0.6} = 2.00, \quad p \approx 2P(Z \geq 2.00) = 0.0456.$$

- At $\alpha = 0.10$: $0.0456 < 0.10$ so reject H_0 .
- At $\alpha = 0.05$: $0.0456 < 0.05$ so reject H_0 .

With stricter quality escalation $\alpha = 0.01$, $0.0456 > 0.01$, so fail to reject H_0 .

Conclusion: The alternative remains “different” throughout; only the significance level changes the decision threshold.

8 App response time: lower vs different on the same data

A software team completed an optimization update and wants to know whether the system is now faster than the response time target promised to users. The benchmark is two hundred fifty milliseconds, so the first claim asks whether the true average has moved lower than that standard. A separate review group asks a broader question about whether response time has changed at all from the same benchmark. Using one common sample, engineers measured forty nine response observations with an average of two hundred forty six point eight milliseconds. Prior performance studies support using eight milliseconds as the known standard deviation, and the analysis uses a five percent significance level.

Questions/Tasks.

(C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) **(Extra analysis)** Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Mean setting with $\mu_0 = 250$, $n = 49$, $\bar{x} = 246.8$, $\sigma = 8$, $\alpha = 0.05$.

C2

Formulation A (lower). $H_0 : \mu = 250$, $H_a : \mu < 250$, left-tailed.

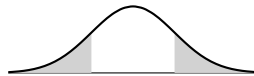
Formulation B (different). $H_0 : \mu = 250$, $H_a : \mu \neq 250$, two-tailed.

C3

Formulation A (lower). Left-tailed test.



Formulation B (different). Two-tailed test.



Extra analysis

$$z = \frac{246.8 - 250}{8/\sqrt{49}} = \frac{-3.2}{1.1429} \approx -2.80.$$

$p_A \approx P(Z \leq -2.80) = 0.0026$. Since $0.0026 < 0.05$, reject H_0 .

$p_B \approx 2P(Z \leq -2.80) = 0.0052$. Since $0.0052 < 0.05$, reject H_0 .

Comparison conclusion: Both tests reject H_0 at 5%, but the two-tailed p-value is larger because evidence is split across two directions.

9 Daily study-time claim: higher vs different on the same data

An academic coach introduced a planning workshop and wants evidence that students are spending more time studying each day than the program baseline. The benchmark is two point zero hours per day, and the coach cares about improvement above that level because the workshop was designed to increase study time. A second reviewer asks a broader question about any change from the same baseline, whether higher or lower. From one shared sample

of thirty six students, the recorded average was two point three two hours per day. Existing records support a known standard deviation of one point two hours, and both analyses use a five percent significance level.

Questions/Tasks.

(C1) Identify parameter type and symbols.

(C2) Write the hypothesis pair that fits the context.

(C3) Classify the tail and justify it.

(EA) (**Extra analysis**) Compute the test statistic, estimate the p-value, compare with α , and state the conclusion in context.

C1

Mean setting with $\mu_0 = 2.0$, $n = 36$, $\bar{x} = 2.32$, $\sigma = 1.2$, $\alpha = 0.05$.

C2

Formulation A (higher). $H_0 : \mu = 2.0$, $H_a : \mu > 2.0$, right-tailed.

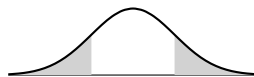
Formulation B (different). $H_0 : \mu = 2.0$, $H_a : \mu \neq 2.0$, two-tailed.

C3

Formulation A (higher). Right-tailed test.



Formulation B (different). Two-tailed test.



Extra analysis

$$z = \frac{2.32 - 2.0}{1.2/6} = \frac{0.32}{0.20} = 1.60.$$

$p_A \approx P(Z \geq 1.60) = 0.0548$. Since $0.0548 > 0.05$, fail to reject H_0 .

$p_B \approx 2P(Z \geq 1.60) = 0.1096$. Since $0.1096 > 0.05$, fail to reject H_0 .

Comparison conclusion: Both formulations fail to reject H_0 at 5%. The two-tailed version is less significant because it tests departures in both directions.